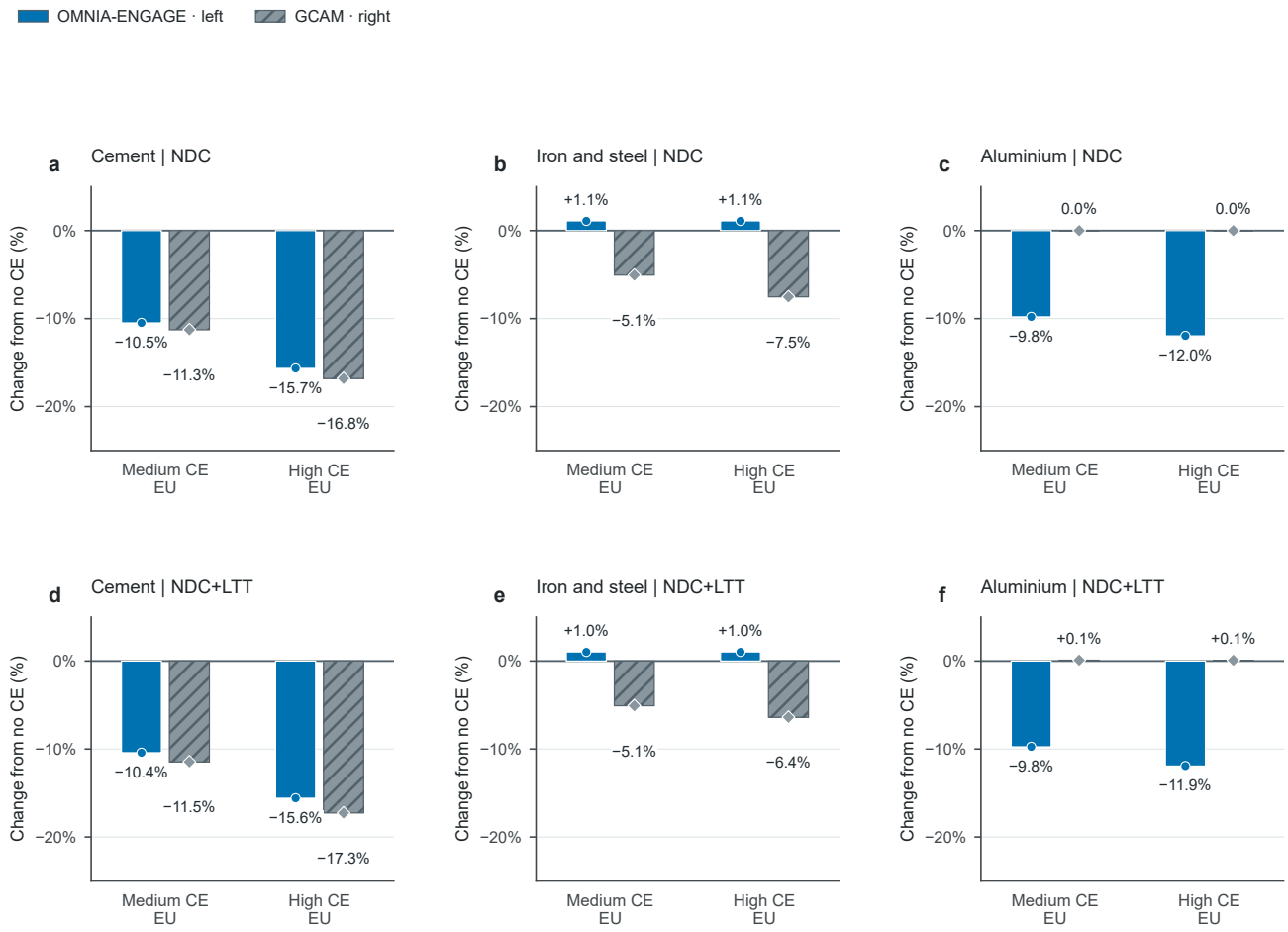
Circular-economy effects on European material production

EU-only CE in 2050 | change from each model's own 2050 no-CE case



Reference: each model's own 2050 no-CE case in the same pathway and sector (0%).

Bars and labels: total output change. Cement excludes clinker; OMNIA metals include both production routes.

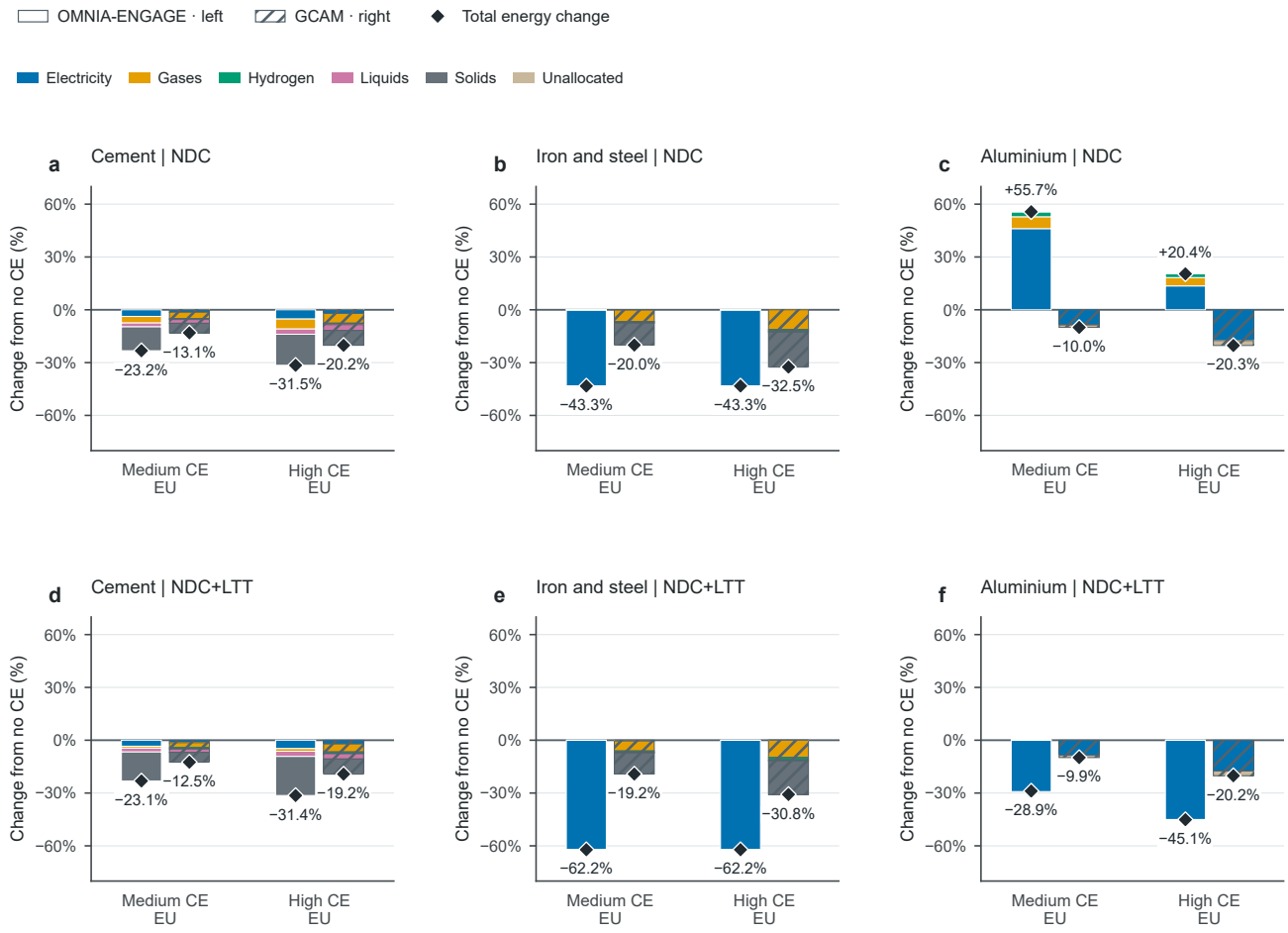
Aluminium: GCAM production-route coverage is unconfirmed. Labels rounded to 0.1%.

Within-model normalisation; no absolute-level alignment assumed. EU-region boundaries differ slightly.

OMNIA missing activity = zero (provisional); GCAM missing values remain gaps.

Circular-economy effects on European final-energy demand

EU-only CE in 2050 | change from each model's own 2050 no-CE case



Reference: each model's own 2050 no-CE case in the same pathway and sector (0%).

Fuel segments: contributions in percentage points. Diamonds and labels: total energy change (%).

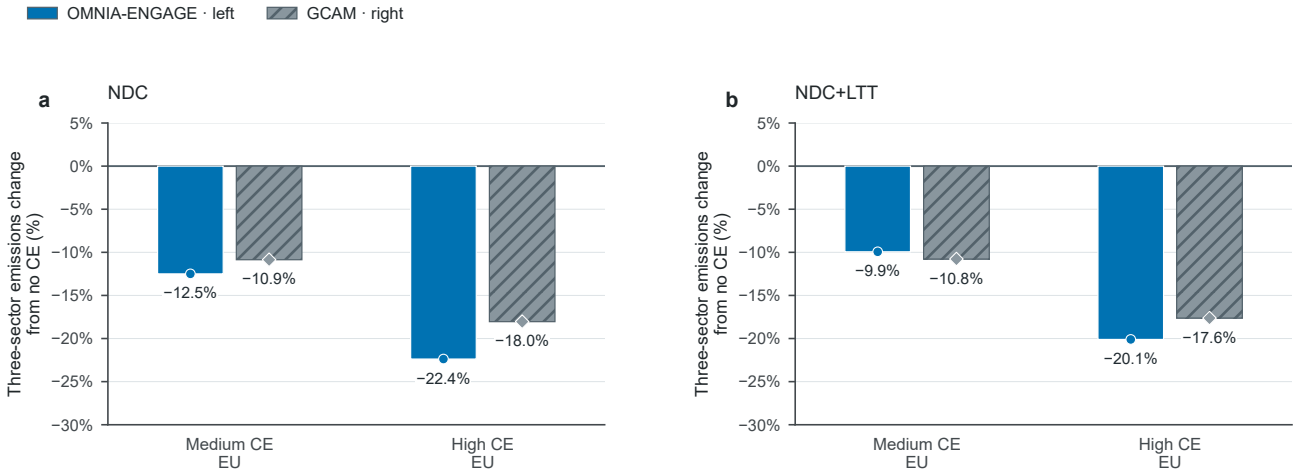
Aluminium: GCAM alumina-refining coverage is unconfirmed; Unallocated retains unidentified energy.

Within-model normalisation; no absolute-level alignment assumed. EU-region boundaries differ slightly.

OMNIA missing activity = zero (provisional); GCAM missing values remain gaps.

Circular economy effects on European material-sector emissions

EU-only CE in 2050 | cement + steel + aluminium | change from own no-CE case



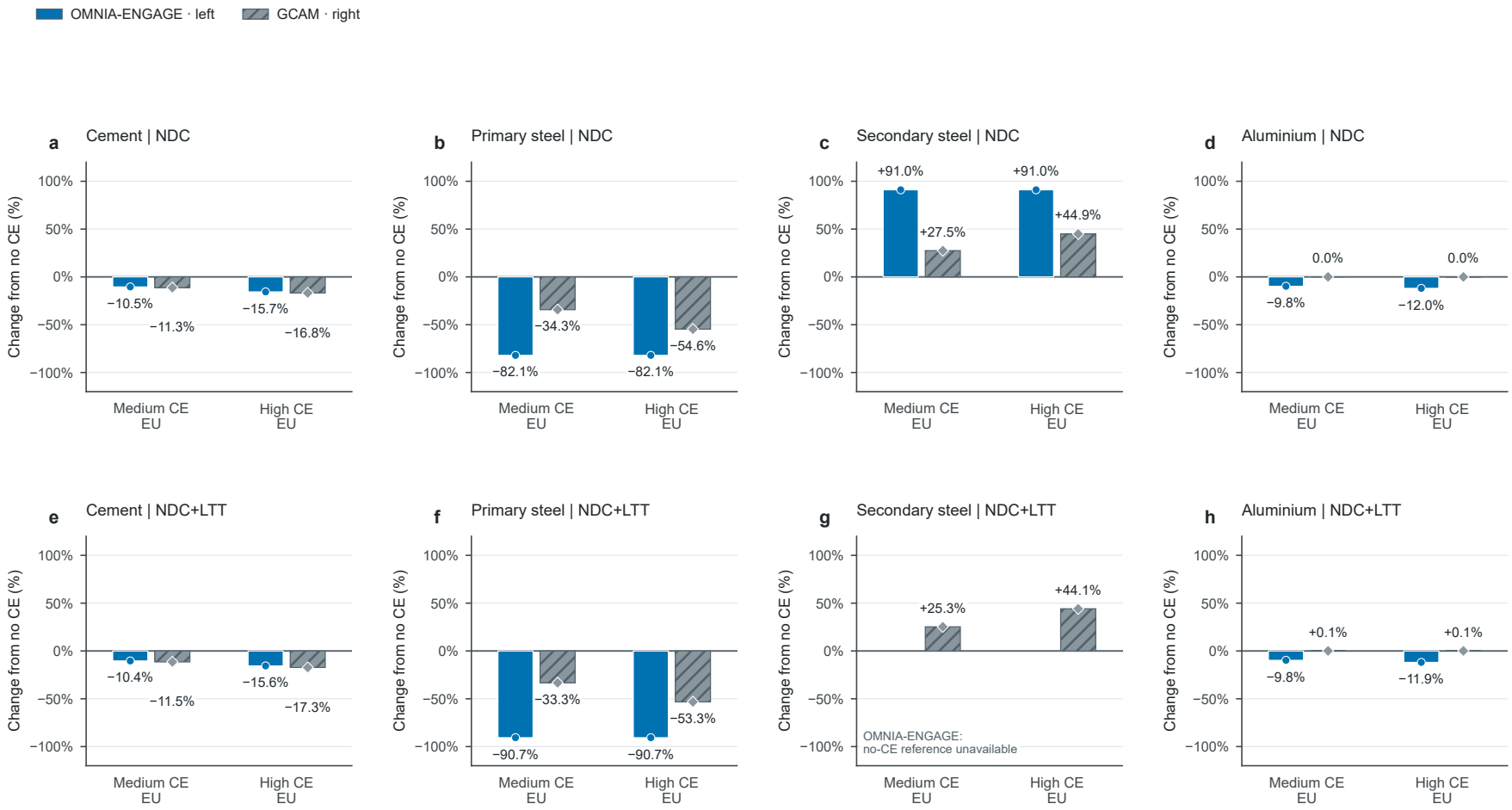
Combined cement, steel and aluminium: OMNIA GHG; GCAM CO2. Gas and process boundaries differ.

GCAM: cement energy + processes; steel/aluminium energy CO2. Reference: own 2050 three-sector no CE.

NDC+LTT aluminium blanks in OMNIA = zero (provisional). Capture is not subtracted again.

Circular-economy effects on European production: separate steel routes

EU-only CE in 2050 | change from each model's own 2050 no-CE case



Reference: each model's own 2050 no-CE output in the same pathway and production category (0%).

Primary and secondary steel each use their own route reference. These percentage changes are not additive.

NDC+LTT secondary steel: OMNIA no-CE inputs are missing; its percentage response is undefined.

Cement and aluminium retain Figure 1's total-output definitions. GCAM aluminium route coverage remains unconfirmed.

All eight panels share one percentage scale. Labels rounded to 0.1%; EU-region boundaries differ slightly.

OMNIA missing additive activity = zero (provisional); GCAM missing values remain gaps.